Aman Shahare

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.90
Dharampeth M.P. Deo Memorial Science College, Nagpur	Class XII (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2018	58.77%
Spick And Span Convent, Nagpur	Class X (Maharashtra State Board of Secondary and Higher Secondary Education, Pune)	2016	77.00%

POSITIONS OF RESPONSIBILITY

Teaching Assistant -- Engineering & Solid Mechanics

Handled tutorial sessions, assignment grading, and doubt-clearing for undergraduate mechanics courses.

Evaluated weekly assignments and supported faculty with overall course administration.

Coordinator -- AI4SE Symposium

Coordinated event scheduling, speaker management, venue operations, participant registration and session planning for the AI4SE symposium.

SKILLS

CAE / FEA Tools: ANSYS, Abaqus

CAD Tools: SolidWorks, Creo 7.0

Programming & Scientific Computing: MATLAB, Python, Simulink

RELEVANT COURSES

Robotics, Dynamics & Controls: Soft Robotics, Dynamics and Vibration, Vehicle Dynamics & Modeling, Data Acquisition and Control Lab, Vibration Lab

Solid Mechanics & FEA: Advanced Mechanics of Solids, Finite Element Method, Advanced FEM

Mathematics & Machine Learning: Machine Learning and its Applications, Mathematical Methods for Engineers

Professional Development: Communication Skills: Advanced, Industry Lectures, Seminar

PROJECTS

Design, Control Architecture, and Hardware Implementation of a Custom Single-Channel ABS

M.Tech Projects

- **Custom Single-Channel ABS - Vehicle Dynamics Project:** Designed and implemented a prototype ABS control

system integrating Hall-effect wheel-speed sensing, real-time speed estimation, hydraulic solenoid control, and hardware fail-safe protection for experimental braking studies.
- Developed a real-time speed measurement and control architecture using **Arduino Mega 2560, Hall sensors, LM393 comparators, and hydraulic actuators**, with testing focused on wheel-speed response and solenoid activation.
- Investigated **tire slip, wheel-lock behavior, sensor fusion, and ABS control logic** to support experimental evaluation of braking performance.

Thermomechanical Modelling of LPBF Cantilever using FEM

M.Tech Course Project | IIT Hyderabad

- Studied how **cooling and substrate preheating affect residual stress and deformation** in an Inconel 718 LPBF cantilever.
- Built and analyzed the model in **Abaqus**, using thermal loading and realistic fixed-end constraints to simulate the final cooling stage.
- Compared different cooling conditions and evaluated **Von Mises stress and vertical displacement** to understand the effect of preheating.
- Found that **substrate preheating significantly reduces distortion and residual stress**, improving the dimensional stability of the printed component.

Machine Learning and Its Applications

M.Tech Course Project | IIT Hyderabad

- Implemented **regression, classification, ANN, ensemble, CNN, and transfer-learning models** using Python, scikit-learn, XGBoost, and PyTorch.
- Applied ML to **engineering and automotive datasets**, including iron-alloy mechanical-property prediction, EV material selection, and automotive sensor-data classification/regression. - Performed **data preprocessing, feature engineering, model training, optimization, hyperparameter tuning, and performance evaluation** using RMSE, MAE, R^2, accuracy, precision, recall, and F1-score.

Autonomous Vehicle Guidance and Control Systems

M.Tech Course Project | IIT Hyderabad

- Developed an autonomous vehicle simulation using **SolidWorks, MATLAB/Simulink**, and a **kinematic bicycle model**.
- Implemented **PID longitudinal control** and **Stanley lateral control** for velocity and path tracking.
- Simulated a **600 m hybrid trajectory** and achieved stable path tracking with bounded yaw response.

Design and Analysis of Kresling Origami Pattern for Soft Robotics Application

M.Tech Course Project | IIT Hyderabad

- Designed a six-sided Kresling origami structure using CAD for soft robotics applications.
- Performed nonlinear finite element analysis in **Abaqus** under axial compression and pressure loading.
- Compared **Neo-Hookean, Mooney-Rivlin, and Ogden** hyperelastic material models based on deformation and von Mises stress.
- Implemented **self-contact, NLGEOM, and C3D4H hybrid tetrahedral meshing** to model large-deformation behavior.
- Analyzed material-dependent structural response, identifying differences in stiffness, stress, and large-strain deformation.

B.E. Final-Year Project

Design and Fabrication of Wheel-Operated Pesticide Sprayer

- Designed and fabricated a manually operated pesticide spraying system for agricultural applications.
- Developed a wheel-driven mechanism to power the pump for continuous spraying.
- Ensured efficient pesticide distribution at controlled pressure and height.
- Focused on cost-effective and user-friendly design for rural applications.

EXTRACURRICULAR

Presented **"Motorcycle Hydraulic ABS Brake Mapping & Slip Control"** at **MathWorks Community Day, Hyderabad**, showcasing MATLAB-based ABS modelling, simulation, and slip-control design.